A Head Camera Is (Almost) All You Need

An electronics-free UMI glove labelled and observed by one moving camera

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github.com/YosubShin/head-cam-umi

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Why a head camera only

- **What the operator sees is what the policy sees**: no privileged observation that only the human had.
- **No synchronization**: one frame is the observation and carries all three labels: camera pose, gripper pose, jaw opening. No cameras, IMUs or encoders to align.
- **Electronics-free glove**: cheap, robust, and scalable.

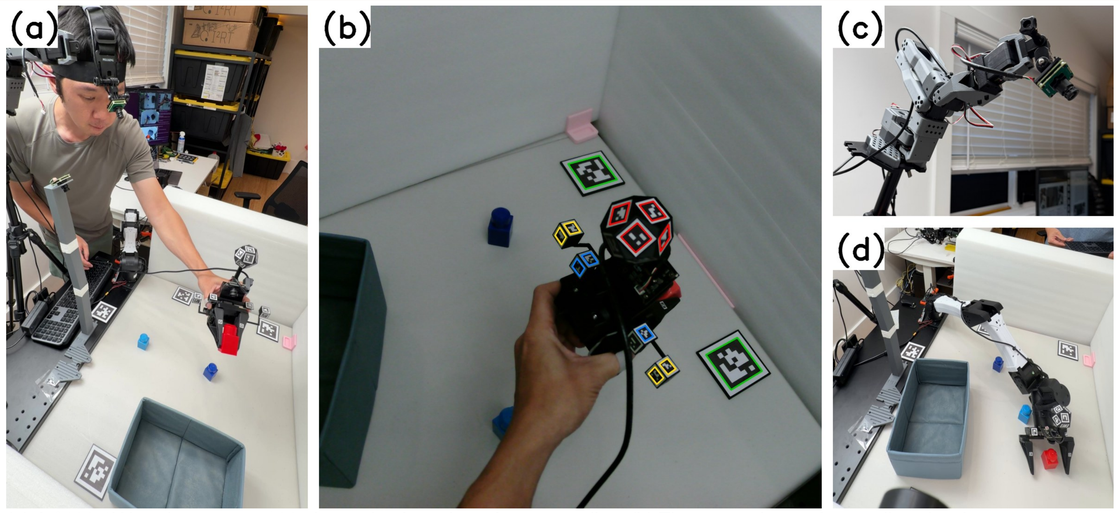
$22 glove, printed

370 g on the hand; UMI 780 g

$122 head camera

103 g on the head

One head camera: the policy's observation and the pose labels



(a) A $122 4K head camera, a $22 electronics-free glove, four paper tiles. (b) The camera's own frame: tiles give the camera pose, the ball the gripper pose, the jaw tags the opening. (c) At deployment the same camera rides a $122 six-joint arm (d) beside the robot.

Label accuracy

- **Camera pose** from the paper tiles.
- **Gripper pose** from ArUco tags on a dodecahedron ball.
- **Jaw width** from printed tails on the jaws.

6.5 mm tool point · **1.6°** orientation · **3.5 %** jaw width, of stroke

Median error against the robot's encoders; a label on 93–100 % of frames.

Collection

- **The operator wears the head camera**; the glove has no wrist camera, IMU or encoder.
- **No special protocol**, such as keeping the wrist facing the bin: what the operator sees is what the policy sees.

2–3× demos per minute vs teleop · **6–8 s** per demo; teleop 14 s

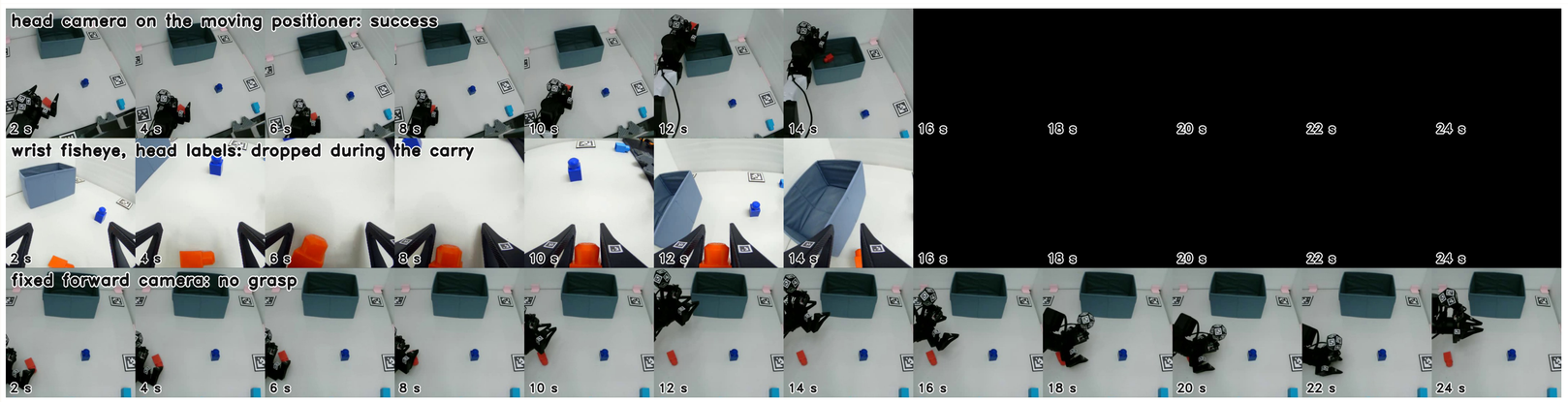
Continuous recording, episodes cut by a keypress.

Deployment

- **Same 4K camera on a modified SO-101 arm**: gripper joint replaced by the camera; $122 in parts.
- **A simple linear model** of the collector's head motion relative to the gripper drives it from the robot's encoders alone.
- **No wrist camera** necessary.

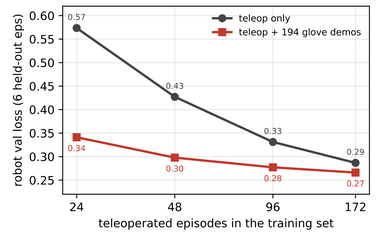
46 mm head pose vs the collector · **5.4°** static camera: 87 mm, 7.0°

One placement, seen by each policy



Placement 23, each policy's own frames every 2 s on one clock, black once its episode ended. Head camera: gripper and bin in one frame through the carry, done in 16 s. Wrist fisheye: only block and wall after the grasp; drops the block before reaching the bin. Fixed camera: sees the whole scene but never closes the gripper (41 s, cut here).

Cheap glove demos complement a small teleop set



10/25
48 teleop episodes

17/25
48 teleop + 194 glove demos

1 ≈ 2–3
teleop episode ≈ glove demos

9 placements won to 2 lost ($p = 0.065$). On the curve (head-label chain, own held-out targets) the 194 glove demos are worth **65–100 teleop episodes**. With all 172 they still help: 18/25 → 20/25 (4 won to 2 lost, $p = 0.69$).

Head-camera labels can replace the encoders

teleop labelled by	encoders	head camera
validation loss	0.318	0.328 (+3 %)
rollouts	18/25	15/25 ($p = 0.58$)

Same 172 teleop episodes, same held-out encoder targets; 5 placements won to 8 lost ($p = 0.58$).

Where each view fails

failures of 25	no grasp	dropped	success
head camera	**5**	0	20
wrist fisheye (UMI)	6	**4**	15
wrist fisheye, head labels	3	**8**	14
fixed camera	**12**	1	12

Wrist policies **drop the block mid-carry**, always while the fisheye looks away from the bin. Head and fixed cameras keep the bin in frame and fail only by not grasping.

It is the view, not the labels

Same view, other labels

wrist fisheye observation

glove labels from	success	dropped
fixed camera	15/25	4
head camera	14/25	8

Same labels, other view

head-camera glove labels

observation	success	dropped
wrist fisheye	14/25	8
head camera	**20/25**	**0**

Limits and next

Limits

- One task, one collector, one tile layout.
- Teleop watched the head-camera view.
- Glove-only policies do not transfer; co-training is required.

Next

- Precision tasks, e.g. cable ties.
- Bimanual, with a second glove.
- Tile-free sessions in the wild.

Setup: eight ACT policies, joint-space actions. Rollouts: 25 placements, one trial each, 60 s. Every co-trained mix uses the same 194 glove demos.